

Wuxi YuHuang Transmission Technology Co., Ltd

无锡羽凰传动科技有限公司



YH8-523D

[Linear Actuator]

[Contact]

Email: imdadul@yhtransmissiontech.com

Phone or WhatsApp: +86-15895307823

Address: 235-Xinzhou Road, Xinwu
District, Wuxi City, Jiangsu, China

YH-523D

Product Overview

1. DC brushed motor
2. The stroke can be customized from 20mm to 500mm;
3. Speed 5-80mm/s optional;
4. Maximum load 5,000N;
5. Operating temperature: -40-65
6. Protection class IP65;



Learn how to choose

S:stroke

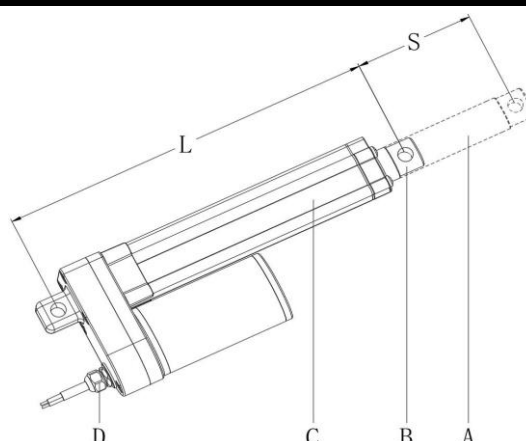
L:Retracted Length

A:Stainless steel inner tube

B:Galvanizednickel connector

C:Aluminum alloy tube body

D:waterproof cable connector



Stroke-	The maximumdistanceapushrodcanmove,thatis,thesizeoffullextension
Retracted length	-minusthesizeoffullretraction.[Customizable] Size when the club is fully retracted. [Customizable]
Front Connector	Multiple forms are available.
Rear Connector	Multiple forms are available.
Install Hole	0° or 90° optional.
Dynamicload	The maximum load that a push rod can withstand while in motion.
Self-locking	The maximum load that a push rod can withstand when it is stationary.
force of the end connection installation hole	IPXX: The first X means dustproof, the second X means waterproof, and the higher the number, the better the protection level. After a minute of continuous operation, you need to rest b minutes, and the duty cycle is the percentage of a/ (a+b).
Protection level	Including no-load speed and full load speed.
Duty Cycle	The output voltage pulse is used to measure the displacement by counting the pulses, and the direction of the push rod can be identified by using the phase difference of the waveform.
Speed	
Hall sensor	

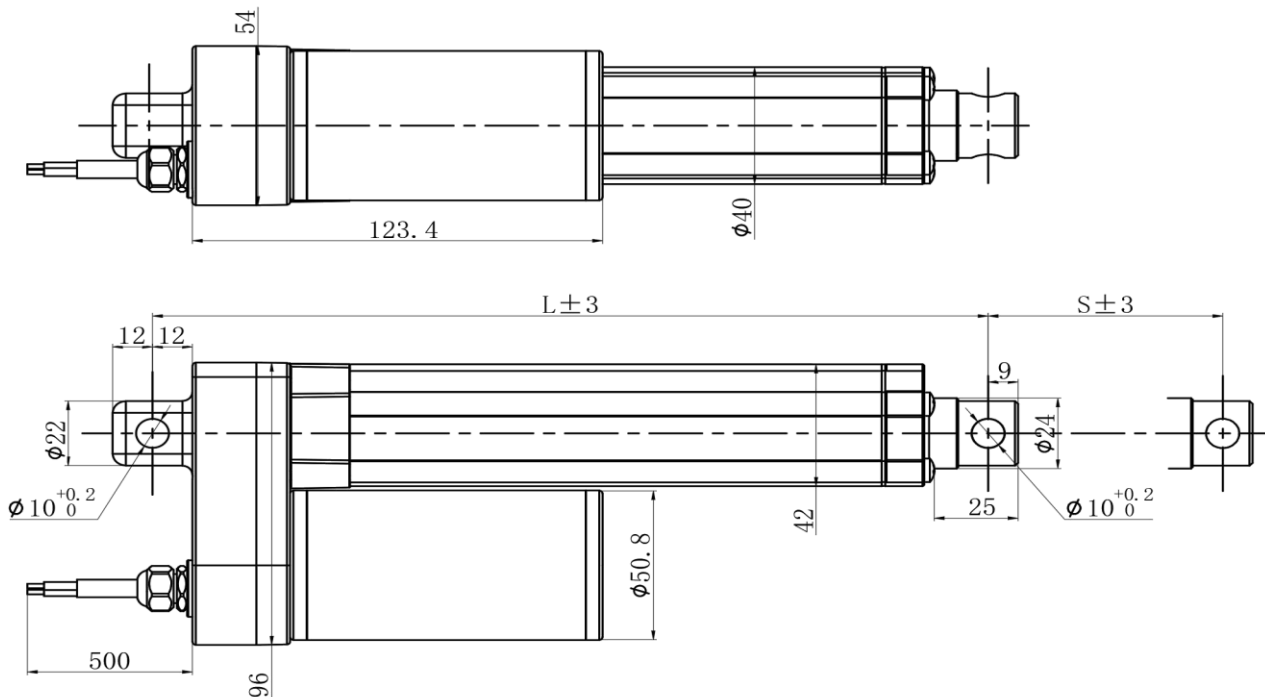
Selection code	523D	X	X	X	X	X	X	X	X	X
Type of club										
YH-523D selection code: 523D										
voltage										
12V	1									
24V	2									
36V	3									
48V	4									
Load and speed										
Refer to Table 1.0										
route or distance of travel										
Refer to Table 4.0										
installation size										
Refer to Table 4.0										
Front connector form										
8.2 mm hole					1					
10.2 mm hole									2	
12.2 mm hole									3	
. 8.2 mm hole with 6.2 mm U-shaped groove									4	
. 10.2 mm hole with 6.2 mm U-shaped groove									5	
M 10x 1.5x20L external thread									6	
M 10x 1.5x20L internal thread									7	
12 rod end joint bearing									8	
Tail end joint form										
8.2 mm hole					1					
10.2 mm hole									2	
12.2 mm hole									3	
. 8.2 mm hole-90°									4	
10.2 mm hole-90°									5	
12.2 mm hole-90°									6	
power line										
0.6 m power cable									1	
1.0 m power cable									2	
1.5 m power cable									3	
2.0 m power cable									4	
customization										5
Waterproof function										
IP65 (standard) Signal output No signal output										0
Limit switch signal lead out										1
Hall signal										2
Electrode signal output										3
ball screw										2

Table 1.0 : Load and Speed Reference Table

code	Max Load Force	Self-Locking	Reduction Ratio	flight lead	No load speed	load speed	12v load current	24v load current	36v load current	48v load current	Max Stroke
	(N)	(N)	-	(mm)	(mm/s)	(mm/s)	(A)	(A)	(A)	(A)	mm
A	5000	7000	40: 1	3	5.5	4.0	9.0	4.5	3.0	2.5	1000
B	4500	7000	40: 1	4	7.2	6.0	9.0	4.5	3.0	2.5	1000
C	1500	2000	40: 1	9	18	14	7.4	3.7	2.5	2.0	1000
D	4000	6000	30: 1	3	7.2	6.0	9.0	4.5	3.0	2.5	1000
E	3000	5000	30: 1	4	9.2	7.0	9.0	4.5	3.0	2.5	1000
F	2500	4000	30: 1	6	13.5	10	9.0	4.5	3.0	2.5	1000
G	1000	1500	30: 1	9	20	16	7.0	3.5	2.5	2.0	1000
H	300	500	10: 1	6	36	32	7.0	3.5	2.5	2.0	1000
I	100	200	10: 1	9	60	50	7.0	3.5	2.5	2.0	1000
J	500	800	5: 1	3	38	33	9.0	4.5	3.0	2.5	1000
K	50	100	5: 1	9	100	80	4.0	2.0	1.5	1.0	1000

1. The data measurement is under the condition of ambient temperature 20 and stable voltage.
2. For example, when the load is 3900N, (D) gear has met the load requirements. Of course, you can also choose (B) or (A) gear to obtain higher load margin and safety factor, as well as longer product life.
3. There are many factors that affect the 'maximum customizable stroke', such as load, speed, direction of force, frequency of use, etc., so it needs to be combined with there a application scenario. If the parameters you need are beyond the optional range, please contact us for further evaluation.

Table 4.0 : The stroke and retracted length



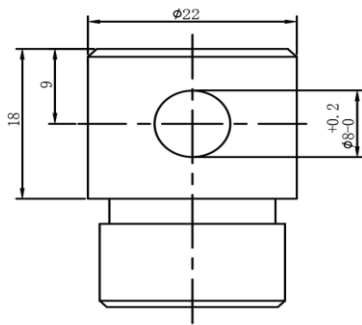
Note: 1>.20Stroke200,L=S+130

200<Stroke≤600,L=S+150

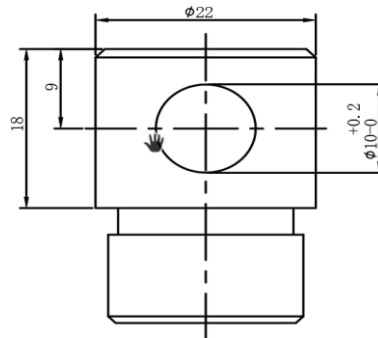
2>.The holedirection in the figure above is 0°

3>.When the front end joint is a rod end joint bearing or a customized joint, the installation size is not applicable to the above rules

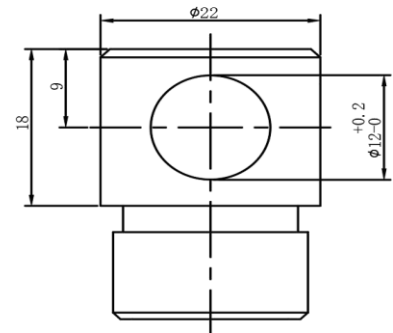
Front Connector



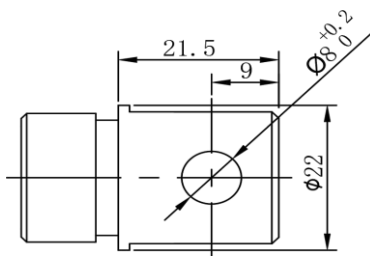
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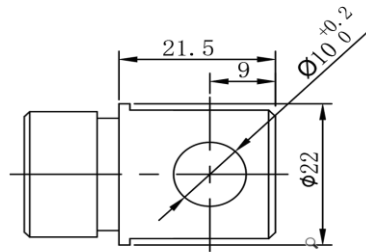
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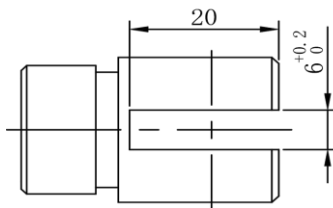
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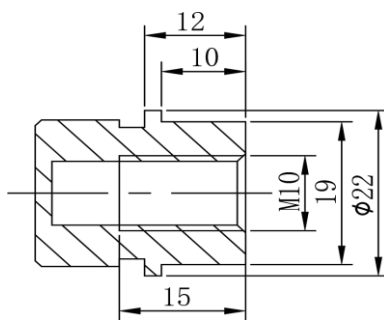
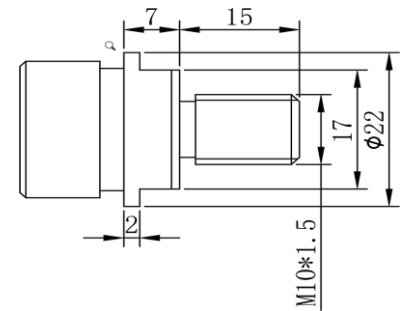
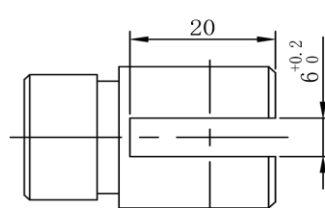
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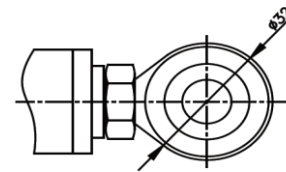
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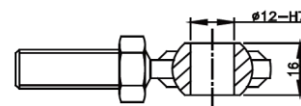
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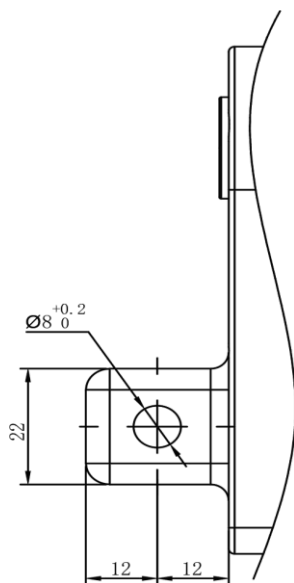
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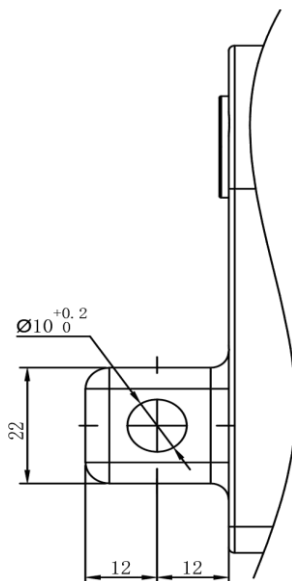
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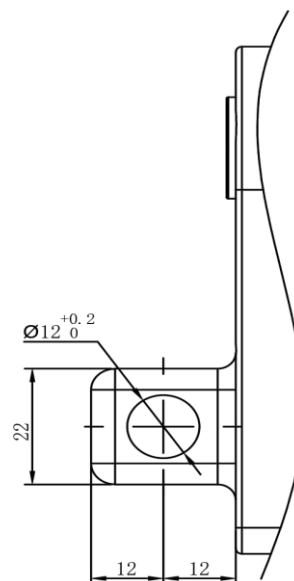
Rear Connector



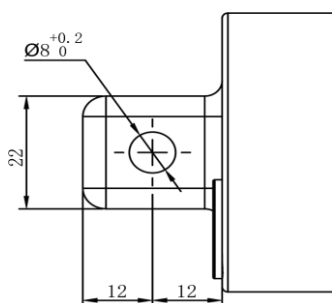
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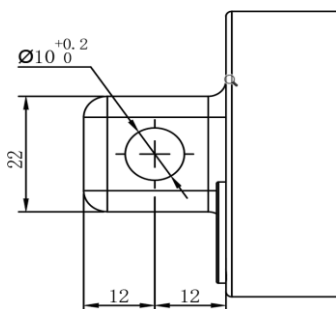
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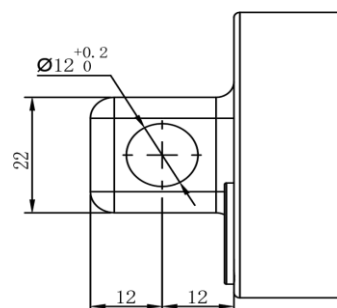
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04



05

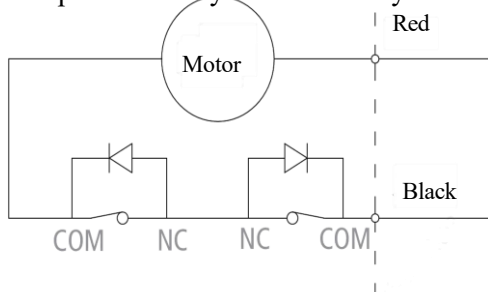


06

Signal Output

0. Built-in limit switch, no signal feedback

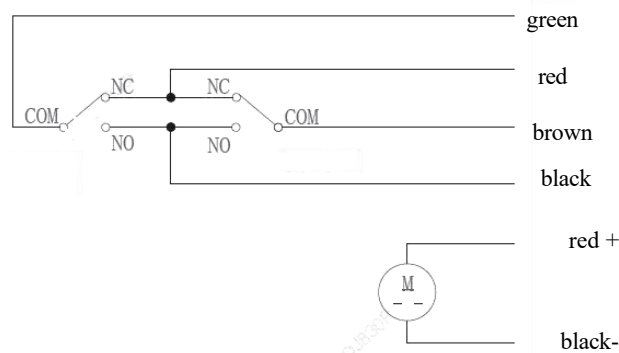
The pushrod has a default upper limit switch and a lower limit switch. When the pushrod is fully extended or fully retracted, the limit switch automatically cuts off the power supply.



Telescopic action wiring method		
color	black	red
extend	-	+
retract	+	-

1. Upper and lower limit signals are external

When the pushrod is fully extended or fully retracted, the limit switch will not automatically cut off the power supply and needs to be controlled externally.

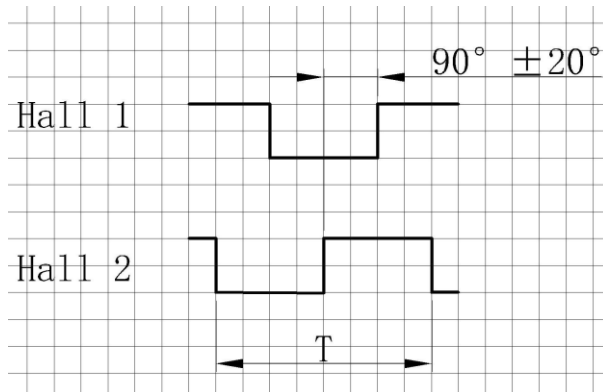


Telescopic action wiring method		
color	black	red
extend	-	+
retract	+	-
signal wire		
black	up, down limit NO	
red	up, down limit NC	
green	up limit COM	
brown	down limit COM	

2. Hall signal feedback

code	feedback	pole quantity	hall pulse/mm
A	hall sensor	4 poles	49.31
B			36.98
C			16.44
D			38.33
E			28.75
F			19.17
G			12.78
H			6.98
I			4.65
J			7.07
K			2.36

* The number of pulse signals corresponding to each 1mm travel of the Hall



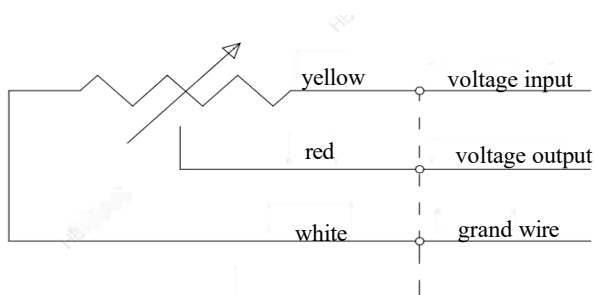
* Hall pulse signal

output waveform diagram

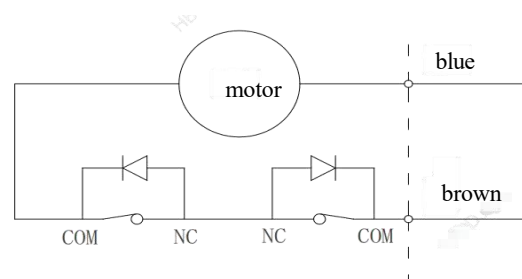
3. Potentiometer signal

feedback

code	7.8:1 reduction ratio max stroke	The resistance value corresponding to stroke	9.8:1 reduction ratio max stroke	The resistance value corresponding to stroke	14:1 reduction ratio max stroke	The resistance value corresponding to stroke
A,D,J	230	0.043	290	0.034	420	0.024
B,E	310	0.032	390	0.026	560	0.018
F,H	460	0.022	580	0.017	840	0.012
C,G,I,K	700	0.014	880	0.011	1260	0.008



potentiometer wire connection



potentiometer signal motor wire connection